

RAPHAEL DUCHET

Master's student in Computer Science (Video Games), UQAC (Chicoutimi, Québec, Canada)
Gameplay & Engine Programmer, Tech Artist, Game & Narrative Designer

Originally from Reunion Island (France)

CONTACTS



+33 7 66 60 90 37



rd.pierrepoivre@gmail.com



[Raphaël D](#)



[WLBib](#)



PORTFOLIO

<https://raphduc.github.io/portfolio-first-iteration/>

PERSONAL QUALITIES

Teamwork, autonomy, perseverance, curiosity, eagerness to learn, creativity, passion for video games.

SOFTWARE



Game Engines



IDEs



Audio Design



3D Art



Video/Image Editing



2D Art



Narrative Design

LANGUAGES/FRAMEWORKS USED

C++, C#, GDScript, Rust, Python, C, Java, SQL, Firebase, HTML, CSS, PHP, TypeScript, Vue.js, Ada.

GAME DEVELOPMENT SKILLS

Gameplay & Engine Programming : mechanics, game loop, custom physics, online systems (Epic/Steam).

Tech Art : custom shaders and stylized rendering (PS1/N64/Dreamcast, water, cel-shading), VFX, GPU and engine optimization pipelines (HLOD, mesh instancing, Unreal Insights).

Game & Narrative design : level design, mechanics design, dialogue and scenario writing.

2D/3D Art : modeling, texturing, animation basics.

Sound/Music : SFX and ambience creation, music composition.

LANGUAGES



French : Native



English : C1 level



Spanish : B2 level



Japanese : Basic knowledge

INTERESTS

Language travel

Sports, music, literature, tabletop role-playing games

PC Hardware

Theatre

Video games

STUDENT & PERSONAL PROJECTS

Gameplay, Multiplayer Programmer & Tech Artist (Lady Jane, VR Futuristic Submarine (2026), Freezy Time Slider (2023))

- Creation of custom cel-shading shaders with ink lines to improve gameplay readability.
- Level design, 2D/3D art, animations, leaderboard, mechanics design and programming.
- Programming of holographic systems and mesh instantiation optimization via HLOD.

Game & Narrative Designer (Influunt Entis (2024-2026))

- Design of an ARPG game project on Unreal Engine. Narrative research, dialogue writing and artistic direction.
- Full architecture and integration of combat mechanics via the Gameplay Ability System (GAS).
- Design of a custom combo system and AI behaviors.

Engine & AI Programmer (Kill Automaton, Ball Shooter (2026))

- Development of a C++ physics engine integrating a dynamic Broad-Phase Octree ($O(N^2)$ complexity) and impulse-based contact resolution.
- Implementation of a hybrid FSM / Reinforcement Learning AI architecture (PyTorch / ML-Agents).

Retro & Water Rendering Research (Water Shaders (2024),

Pawater, N64 Material, Sours' (2026))

- Development of Materials emulating N64, PS1 and Dreamcast graphics pipelines (color quantization, dithering, distance fog).
- Parametric water shaders (depth, absorption, reflection).
- 3D environment modeling and animated 2D sprites, blend inspired by *Xenogears*.

EDUCATION



Master's in Computer Science (Video Games)
UQAC (Canada) (2025-2026)



Dual Degree : Master's in Video Games/DESS in Narrative Games
HEAJ (Belgium) / NAD-UQAC (Canada) (2025)



Dual Degree : Bachelor's in Video Games/DUETI in Comp. Science
UQAC (Canada)/UGA (France) (2023)

GAMEJAMS



Overdrive Oasis : jet-ski and shader research built in 3 hours
TriJam#375, 06/2026



Death Fel' Ball Tournament : Lethal League meets Super Smash Bros
Montreal Game Jam (Test), 05/2026



Alula : a virtual bird race for up to 4 players
WonderJam Hiver 2025, 02/2025



Hystoripia : an OST created in place of a game
OSTComposingJam#6, 07/2023

INTERNSHIP



Avalon Digital : video game studio, April-June 2022

- Setup of the I2Localization asset and automation of in-game text translation at the launch of [Britannia \(Available on Steam\)](#).



- 3D asset modeling (board game pawns and pieces) in Blender.
- Visual effects and pawn animations, with associated scripts.